

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	Medium

Question

While researching a topic, a student has taken the following notes:

- Planetary scientists classify asteroids based on their composition.
- C-type asteroids are composed primarily of carbon.
- They account for roughly 75 percent of known asteroids.
- S-type asteroids are primarily made up of silicate minerals.
- They account for roughly 17 percent of known asteroids.

The student wants to emphasize a difference between C-type and S-type asteroids. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

- A. Planetary scientists classify asteroids into types, two of which are the C-type and the S-type.
- B. Planetary scientists consider an asteroid’s composition (such as whether the asteroid is composed mainly of silicate minerals or carbon) when classifying it.
- C. Roughly 17 percent of known asteroids are classified as S-type asteroids; another percentage is classified as C-type asteroids.
- D. C-type asteroids are mainly composed of carbon, whereas S-type asteroids are primarily made up of silicate minerals.

Correct Answer: D

Rationale

Choice D is the best answer. The sentence emphasizes a difference between C-type and S-type asteroids, noting that C-type asteroids are mainly composed of carbon, while S-type asteroids are mainly composed of silicate minerals.

Choice A is incorrect. The sentence states that C-type and S-type are two types of asteroids, but it doesn't emphasize a difference between them. Choice B is incorrect because it doesn't directly mention C-type or S-type asteroids. Choice C is incorrect. While the sentence mentions that 17 percent of known asteroids are S-type asteroids, it doesn't identify the percentage of asteroids that are C-type. Therefore, the sentence doesn't emphasize a difference between the two types.